# ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

Hemavathi M

1912222

## C0-3

## EXPERIMENT 10

**Aim**

To verify Faraday's law of electromagnetic induction by determining and plotting the emf induced in a multi-turn loop due to (i) a time-varying magnetic flux density (transformer emf), (ii) conductor motion in a magnetic field (motional emf) and (iii) loop rotation (generator emf), and to confirm the polarity of the induced emf using Lenz's law.

**Test Case (i)**

**Transformer EMF Induced by a Time-Varying Flux Density**

```
clc;

clear;

N  = input('Enter number of turns N: ');

A  = input('Enter loop area A (m^2): ');

B0 = input('Enter peak flux density B0 (T): ');

f  = input('Enter frequency f (Hz): ');

w  = 2*pi*f;

t  = linspace(0,2/f,2001);

dt = t(2)-t(1); B  = B0*sin(w*t);

flux_link = N*A*B;                  % flux linkage (Wb-turn) 1500 rpm emf_num = -gradient(flux_link,dt);
% emf = -d(lambda)/dt (numerical) emf_th  = -N*A*B0*w*cos(w*t);          % analytical emf Em =
N*A*B0*w;

disp('Peak induced emf (V) =');

disp(Em); disp('RMS induced emf (V) =');

disp(Em/sqrt(2));

dev = max(abs(emf_num(2:end-1)-emf_th(2:end-1)))/Em*100;

disp('Maximum deviation (%) =');

disp(dev); subplot(2,1,1);

plot(t*1e3,flux_link,'LineWidth',2);

grid on; xlabel('Time (ms)');

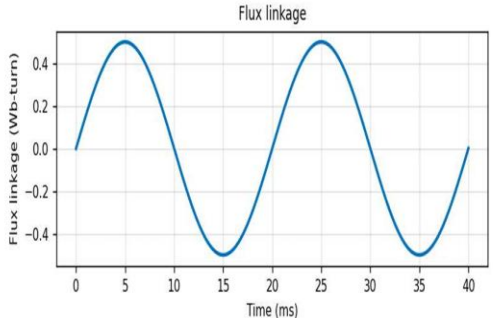
ylabel('Flux linkage (Wb-turn)');

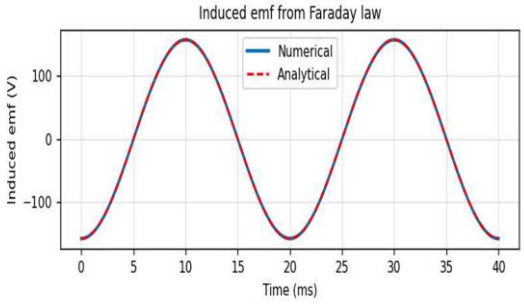
title('Flux linkage');
```

```
subplot(2,1,2);
plot(t*1e3,emf_num,'b',t*1e3,emf_th,'r--','LineWidth',1.5);
grid on;
xlabel('Time (ms)'); y
label('Induced emf (V)');
legend('Numerical','Analytical');
title('Induced emf from Faraday law');
```

**Sample Output:**

For N = 100, A = 0.01 m², B0 = 0.5 T and f = 50 Hz, the peak emf = 157.08 V, the rms emf = 111.07 V and the maximum deviation is below 0.001 %.




**Test Case (ii)**

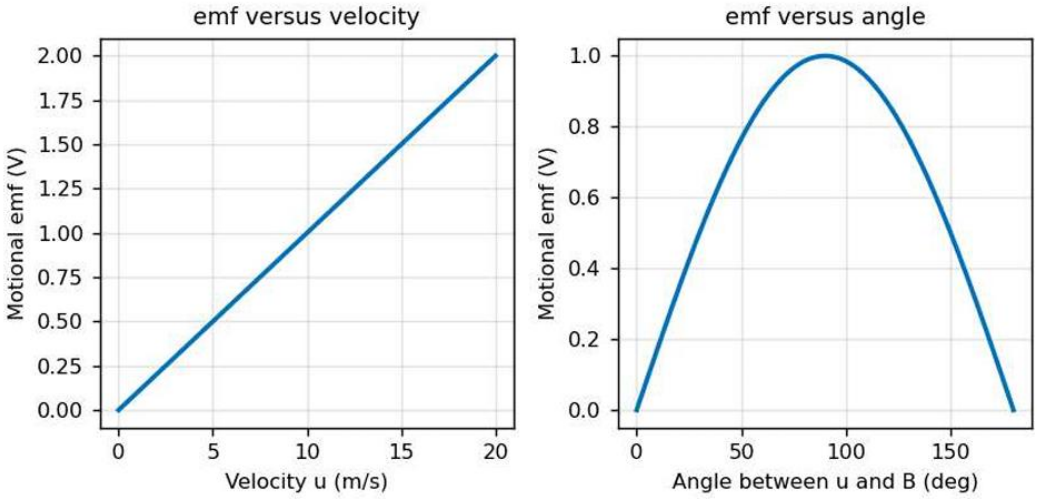
**Motional EMF of a Conductor Moving in a Magnetic Field**

```
clc;
clear;
B  = input('Enter flux density B (T): ');
l  = input('Enter conductor length l (m): ');
u  = input('Enter velocity u (m/s): ');
th = input('Enter angle between u and B (deg): ');
emf = B*l*u*sind(th); disp('Motional emf (V) =');
disp(emf);
uv = linspace(0,20,100);
emf_u = B*l*uv*sind(th);
subplot(1,2,1);
plot(uv,emf_u,'LineWidth',2);
```

```
grid on;
xlabel('Velocity u (m/s)');
ylabel('Motional emf (V)');
title('emf versus velocity');
thv = linspace(0,180,181);
emf_a = B*l*u*sind(thv);
subplot(1,2,2);
plot(thv,emf_a,'LineWidth',2);
grid on; xlabel('Angle between u and B (deg)');
ylabel('Motional emf (V)'); title('emf versus angle');
```

**Sample Output:**

For B = 0.5 T, l = 0.2 m, u = 10 m/s and θ = 90°, the motional emf = 1.00 V; for θ = 30° the emf = 0.50 V.




**Test Case (iii)**

**EMF of a Rotating Loop (Generator Action)**

```
clc;
clear;
N = 100;
B = 0.5;
A = 0.01;
rpm = 1500;
```

```
w = 2*pi*rpm/60;

f = w/(2*pi);

t = linspace(0,2/f,1000);

emf = N*B*A*w*sin(w*t);

Em   = N*B*A*w;

Erms = Em/sqrt(2);

fprintf('Frequency f = %.2f Hz\n', f);

fprintf('Peak emf = %.2f V\n', Em);

fprintf('RMS emf = %.2f V\n', Erms);

 plot(t*1e3,emf,'LineWidth',2);

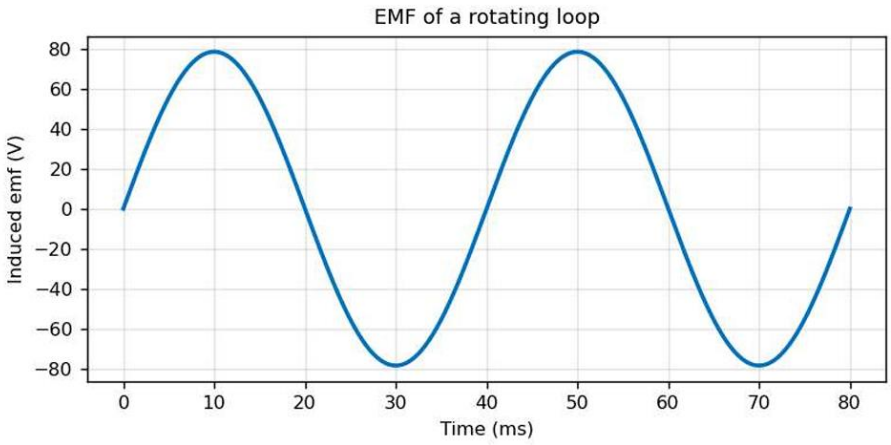
grid on; xlabel('Time (ms)');

ylabel('Induced emf (V)');

title('EMF of a rotating loop');
```

**Sample Output:**

For N = 100, B = 0.5 T, A = 0.01 m² and 1500 rpm, f = 25.00 Hz, the peak emf = 78.54 V and the rms emf = 55.54 V.




**Test Case (iv)**

**Effect of Frequency and Number of Turns on the Peak EMF**

```
clc;

clear;

A = 0.01;

B0 = 0.5;
```

```
f  = linspace(10,100,100);

Nv = [50 100 200]; hold on;

for k = 1:3 Em = Nv(k)*A*B0*2*pi*f;

plot(f,Em,'LineWidth',2);

end grid on;

xlabel('Frequency f (Hz)');

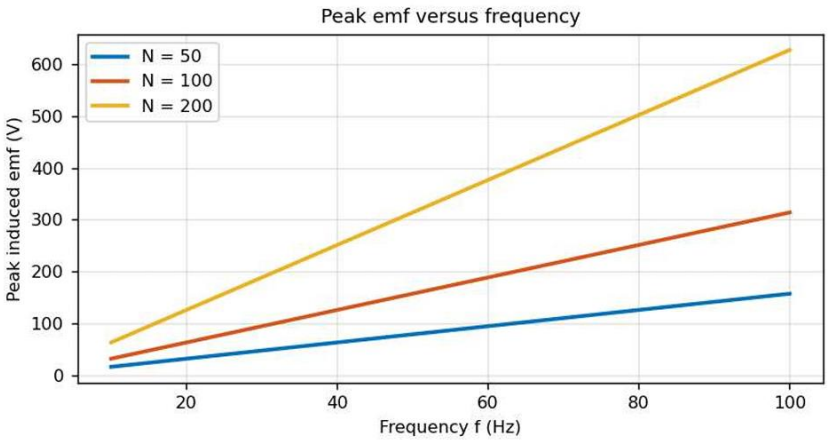
ylabel('Peak induced emf (V)');

title('Peak emf versus frequency');

legend('N = 50','N = 100','N = 200');
```

**Sample Output:**

At f = 50 Hz the peak emf is 78.54 V for N = 50, 157.08 V for N = 100 and 314.16 V for N = 200; for N = 100 at f = 100 Hz it is 314.16 V.




**RESULTS:**

**Thus the Verification of Faraday's Law of Electromagnetic Induction – Determination and Plotting of Induced EMF (Transformer EMF and Motional EMF) have been done using MATLAB software**

| S. No. | Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weighting (%) | Marks given | Marks obtained |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Identify induction mechanism and parameters | Identify induction mechanism and parameters | Correctly identifies flux change/turns for transformer EMF and B, length, velocity/ orientation for | Identifies most parameters correctly. | Some parameters or directions are unclear. | Uses an incorrect physical model. | 25 | 2.5 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | motional EMF. | | | | | | |
| 2 | Calculate induced EMF in MATLAB | Calculate induced EMF in MATLAB | Implements the applicable Faraday and motional EMF equations correctly with units/sign convention. | Calculations are mostly correct with minor errors. | Some calculations require assistance. | EMF calculations are incorrect. | 25 | 2.5 | |
| 3 | Plot and verify EMF behaviour | Plot and verify EMF behaviour | Plots expected variation and verifies results against theoretical relations. | Plots and verification are mostly correct. | Plot or validation is incomplete. | No valid plot or verification. | 30 | 3 | |
| 4 | Interpret and document results | Interpret and document results | Distinguishes transformer and motional EMF and provides clear tables/plots/ conclusion. | Explanation is correct with minor omissions. | Explanation is limited**.** | Results are not explained or documented. | 20 | 2 | |
| Total | | | | | | | 100 | 10 | |